

# ControlPlane.ai

Zero-Trust Guardrail Layer for Enterprise AI | Accenture Innovation Challenge 2026

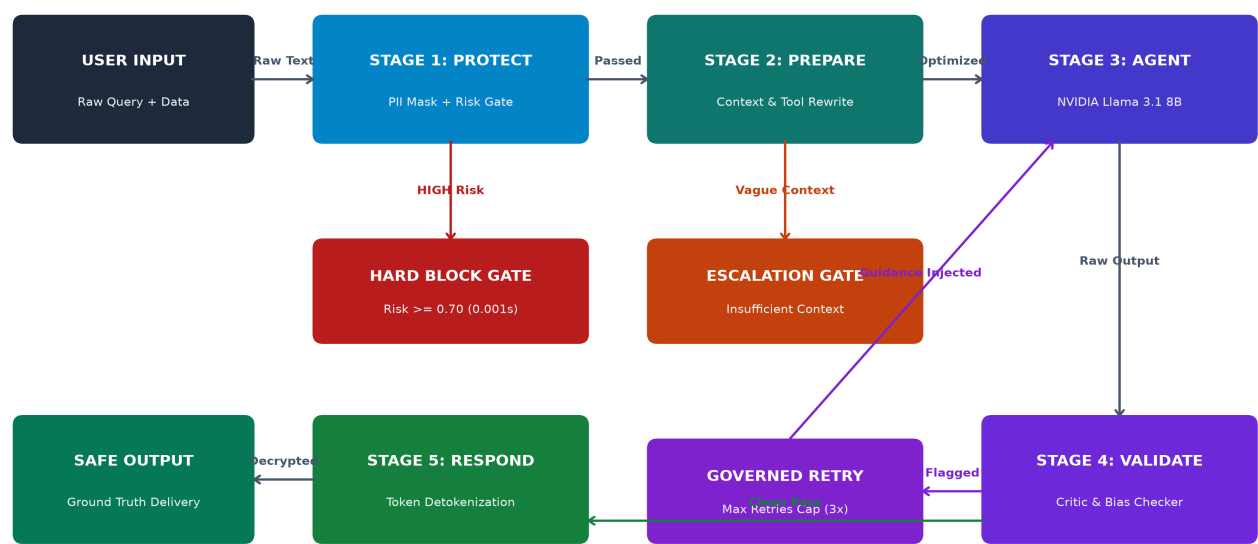
|                                          |                                                 |
|------------------------------------------|-------------------------------------------------|
| Author / Team: Grown Wings               | Target Track: Enterprise AI Governance & Safety |
| Primary Model: NVIDIA Llama 3.1 8B (NIM) | Verified Net Savings: ~52.9% vs Frontier Models |

## 1. Problem Statement & Enterprise Challenges

As organizations transition from passive copilots to autonomous multi-agent systems, they encounter an enterprise AI trilemma: **(1) Privacy & Secret Leaks** (accidental exposure of PII, API tokens, and credentials to third-party model providers), **(2) Factual Hallucinations & Policy Bias** (unverified model assertions and demographic stereotyping causing financial/reputational damage), and **(3) Runaway Compute Costs** (over-reliance on costly frontier LLMs for routine workflows without query optimization).

## 2. Solution Architecture: Observe, Evaluate, Act

ControlPlane.ai is a model-agnostic middleware sitting directly between end users and enterprise agents. It enforces a four-stage pipeline ensuring zero plaintext exposure, deterministic risk gating, and governed validation.



## 3. Stage-by-Stage Operational Breakdown

### Stage 1: Protect (PII / Secret Masking + Risk Gate)

- **Tokenization Engine:** Replaces emails, phones, SSNs, credit cards, and high-entropy secrets (AWS keys, JWTs) with deterministic tokens (<PII\_EMAIL\_1>, <SECRET\_API\_KEY\_1>).
- **Authoritative Risk Gate:** Evaluates prompt injection, jailbreaks, and wire fraud via deterministic rules. Scores >= 0.70 trigger a **HARD BLOCK** in <1ms (\$0.00 model compute).

### Stage 2: Prepare (Context Check + Tool-Aware Rewrite)

- **Input 0 Tool Discovery:** Dynamically discovers available agent capabilities.
- **Context Sufficiency Check:** Identifies vague pronouns (it, that, them) and missing IDs, escalating for human clarification before invoking models.
- **Tool-Aware Query Rewrite:** Strips conversational fluff and injects required tool signatures ([TOOLS: ...]), compressing input tokens.

### Stage 3: Enterprise Agent Inference (NVIDIA NIM Llama 3.1 8B)

- **Model-Agnostic Invocation:** Executes inference via OpenAI-compatible endpoints using Small Language Model (SLM) economics (~\$0.18/1M tokens).
- **Zero-Trust Safety:** The model processes only placeholder tokens, preventing raw credential leakage into provider training datasets.

**Stage 4: Validate (Critic & Bias Checker Agents + Controlled Retry)**

- **Factual Critic Agent:** Dedicated LLM verifying claims against query context to eliminate hallucinations.
- **Bias & Fairness Checker:** Dedicated LLM auditing for stereotyping, unfair generalizations, and safety policy violations.
- **Governed Retry Loop:** If flagged, ControlPlane injects targeted feedback and re-prompts the agent up to MAX\_RETRIES (default: 3).

**Stage 5: Respond (Safe Detokenization & Delivery)**

- **Collision-Free Detokenization:** Restores original plaintext PII strictly after validation passes.
- **Safe Final Output:** Delivers the verified response along with complete audit trail telemetry and latency metrics.

**4. Economic Impact & Net Cost Optimization**

By replacing blind frontier model routing with an optimized SLM agent paired with prompt compression and early risk gate short-circuiting:

| Metric Dimension         | Unconstrained Frontier LLM   | ControlPlane.ai + SLM             | Variance       |
|--------------------------|------------------------------|-----------------------------------|----------------|
| Base Token Pricing       | \$5.00 - \$15.00 / 1M tokens | \$0.18 / 1M tokens (Llama 3.1 8B) | -96.4%         |
| Adversarial Query Cost   | Full inference cost incurred | \$0.00 (Hard-blocked at Stage 1)  | -100%          |
| Prompt Token Consumption | 100% (Uncompressed)          | ~65-75% (Fluff compressed)        | -25-35%        |
| Net Blended Cost         | 100% Baseline                | 47.1% Net Cost                    | ~52.9% Savings |

**5. Execution Guide & Test Suite Verification**

The project is fully tested (58 automated unit & end-to-end tests passing with 98% coverage) and demo-ready.

**Launch Streamlit Dashboard:** `.\controlplane\.venv\Scripts\streamlit.exe run controlplane\streamlit_app.py`

**Run Automated Pytest Suite:** `pytest tests/ -v --cov=controlplane`

**GitHub Repository:** <https://github.com/dhruvkachhela/controlplane>